1 Introduction

The rapid expansion of environmental, social, and governance (ESG) reporting has made corporate environmental disclosure an important information channel for investors, regulators, and other stakeholders. Environmental communication can enhance legitimacy and attract capital, but these benefits also create incentives for firms to present a favorable environmental image without achieving commensurate environmental performance (Delmas and Burbano, 2011). Greenwashing therefore arises when environmental communication creates a strategic divergence from the substantive environmental performance that could support that communication (Lyon and Maxwell, 2011; Zhou et al., 2024). This divergence reduces the credibility and decision usefulness of environmental information because stakeholders may interpret favorable narratives as evidence of environmental conduct that the firm's actual performance does not support (Christensen et al., 2021; Zhou et al., 2024). An important but unresolved question is whether the ordinary financial-statement audit can constrain this communication–performance divergence even when no separate sustainability assurance is provided.

Environmental narratives in annual reports generally fall outside the direct scope of the financial-statement audit opinion. However, they are not necessarily isolated from the audit process. ISA 720 and its Chinese counterpart, CAS 1521, require auditors to read other information and consider whether it is materially inconsistent with the audited financial statements or with knowledge obtained during the audit. These requirements do not make the financial-statement auditor an assurer of environmental information. They nevertheless create a limited institutional channel through which particular environmental statements may be compared with information obtained during the ordinary audit. The possibility of such comparison depends on the content of the environmental statement. General aspirations, policy slogans, and broad commitments usually provide few concrete points of comparison. By contrast, statements describing environmental projects, technology adoption, capital investment, pollution remediation, equipment installation, regulatory rectification, or measurable environmental outcomes may contain factual anchors that have financial, operational, or compliance-related implications (Clarkson et al., 2004, 2008; Barth and McNichols, 1994). Such statements may therefore intersect with information about assets, expenditures, cash flows, production processes, environmental obligations, penalties, or other matters encountered during the audit. Accordingly, concrete action claims are more likely than symbolic claims to provide information that can be assessed with the auditor's existing knowledge of the client and its industry.

Audit firm industry specialization may make this evaluation more informative. Industry-specialist auditors accumulate knowledge about clients' business models, production processes, technologies, regulatory environments, and typical financial relationships (Solomon et al., 1999; Francis, 2011; Reichelt and Wang, 2010). This knowledge can help them recognize when several individually ambiguous signals form an unusual pattern, assess the risk associated with that pattern, and select more targeted follow-up procedures (Owhoso et al., 2002; Low, 2004; Hammersley, 2006). Applied to environmental narratives, industry expertise may improve the probability that an auditor recognizes a concrete environmental claim that is difficult to reconcile with the client's operations, financial information, or information obtained during the audit. If a potential material inconsistency becomes salient, CAS 1521 provides a basis for discussing the matter with management, considering whether further work is necessary, and communicating an uncorrected material misstatement to those charged with governance. This process may discourage environmental action claims that management cannot adequately support or reconcile with the firm's audited financial, operational, and compliance-related information. To the extent that such unsupported claims

inflate the substantive appearance of environmental disclosure relative to the firm's underlying environmental performance, their reduction narrows the gap captured by our measure of greenwashing. This reasoning leads to our research question: Is audit firm industry specialization associated with less corporate greenwashing?

We examine this question using 15,381 firm-year observations for Chinese A-share listed firms from 2021 through 2024. We view greenwashing as the use of environmental language to create a favorable environmental image without corresponding substantive environmental improvement. Accordingly, we measure greenwashing as the within-industry-year divergence between a disclosure-side measure and an external environmental-performance score. Following the DeepGreen procedure (Xu et al., 2026), we use LLM-enabled semantic classification of annual-report text to distinguish substantive statements describing firm-specific environmental actions from symbolic statements such as aspirations, slogans, and general commitments. The resulting green implementation ratio (GI) is the share of classified environmental statements that are substantive. A GI approaching one indicates that the firm's identified environmental communication is predominantly supported by descriptions of concrete actions or improvements; a GI approaching zero indicates that it is predominantly symbolic. GI therefore captures the substantive effectiveness of the disclosure side, rather than the sheer amount of green language. Conventional measures based on environmental word counts, statement counts, report length, or textual tone primarily capture disclosure volume or rhetorical positivity; they combine substantive and symbolic communication and may increase when firms accumulate generic green language to project a favorable image (Lublóy et al., 2025; Yu et al., 2020; Grewal et al., 2022; Walker and Wan, 2012). In contrast, the total number of classified statements captures disclosure quantity, while GI distinguishes the substantive component from that quantity by identifying whether the communication describes a concrete environmental action (Clarkson et al., 2008; Hummel and Schlick, 2016; Xu et al., 2026). We identify greenwashing when a relatively high GI is paired with comparatively weak external environmental performance; the resulting communication–performance gap is our greenwashing measure. We measure audit firm industry specialization using the audit firm's share of client operating revenue within an industry-year and a binary indicator of strong specialization equal to one when that share is at least 20 percent (Cai and Xian, 2007; Krishnan, 2003). The analysis controls for financial, governance, and audit characteristics and includes year and industry fixed effects. The main results show that audit firm industry specialization is negatively associated with corporate greenwashing. This negative association is robust across various specifications and robustness tests.

This study contributes to the literature in three ways. First, it extends research on audit firm industry specialization and client disclosure quality by examining whether industry expertise is associated with the gap between environmental communication and environmental performance (Balsam et al., 2003; Dunn and Mayhew, 2004; Reichelt and Wang, 2010). Second, the study adds to greenwashing research by identifying a potential governance spillover from the ordinary financial-statement audit, rather than focusing only on firms' legitimacy incentives and external environmental scrutiny (Delmas and Burbano, 2011; Lyon and Maxwell, 2011; Marquis et al., 2016). Third, the study uses semantic classification to distinguish substantive from symbolic environmental communication in a large-sample auditing setting (Baier et al., 2020; Bilinski, 2024; Xu et al., 2026).

The remainder of the paper develops the theoretical arguments, explains the measurement and research design, presents the empirical results, and concludes.

2 Literature Review and Hypothesis Development

2.1 Greenwashing as an Action–Performance Divergence

Environmental disclosure can reduce information asymmetry by helping investors and other stakeholders assess firms' environmental responsibilities, exposures, and performance (Healy and Palepu, 2001; Verrecchia, 2001). Its usefulness depends on whether environmental communication provides a credible representation of underlying corporate conduct. When firms use favorable environmental communication strategically, disclosure can obscure rather than clarify environmental performance. Greenwashing is therefore commonly understood as a mismatch between positive environmental communication and the substantive environmental performance that could support that communication (Delmas and Burbano, 2011; Lyon and Maxwell, 2011).

Stakeholder expectations and legitimacy pressures increase the benefits of appearing environmentally responsible, whereas substantive environmental improvements may require costly changes in technology, production, and organizational practices (Delmas and Burbano, 2011). Firms may consequently rely on selective, exaggerated, or symbolic communication when outsiders cannot readily observe the underlying performance. Prior research shows that institutional norms, external scrutiny, and the expected cost of having environmental claims challenged can affect firms' disclosure choices (Kim and Lyon, 2015; Marquis et al., 2016; Lyon and Maxwell, 2011). Taken together, these arguments suggest that greenwashing becomes more attractive when the benefits of a favorable environmental image exceed the expected costs of having unsupported, incomplete, or misleading claims challenged.

According to the measurement of greenwashing, consistent with the greenwashing literature, our core concern is that firms may accumulate environmental-friendly language to construct a favorable image without achieving corresponding environmental improvement (Delmas and Burbano, 2011; Lyon and Maxwell, 2011; Marquis et al., 2016). Conventional green-disclosure measures based on word counts, sentence counts, or overall disclosure volume capture the amount of environmental communication but may not distinguish concrete environmental implementation from symbolic or promotional language (Clarkson et al., 2008; Hummel and Schlick, 2016; Xu et al., 2026). A firm may therefore disclose many environmental words or statements while providing little evidence of concrete action, whereas a smaller number of specific statements may be more informative when they describe actual implementation (Clarkson et al., 2008; Marquis et al., 2016). Conversely, a smaller number of specific statements may be more informative when those statements describe actual implementation. A useful disclosure-side measure must distinguish communication that describes firm-specific environmental action from communication that consists mainly of symbolic expression, and it must then be interpreted jointly with an independent measure of environmental performance.

We use the green implementation ratio (GI)(Xu et al., 2026), defined as the share of classified environmental communication that describes a substantive action rather than a symbolic statement. This choice builds on the distinction between hard and soft environmental disclosure: hard information contains more specific, verifiable, and decision-relevant content, whereas soft information more often consists of policies, principles, aspirations, or favorable descriptions (Clarkson et al., 2008; Hummel and Schlick, 2016). When GI approaches one, most identified environmental disclosure is supported, at least in its content, by descriptions of concrete actions or substantive improvements. When GI approaches zero, the disclosure is dominated by symbolic language or the firm has no classified environmental disclosure. Thus, a higher GI indicates stronger substantive effectiveness on the disclosure side, whereas a lower GI indicates a greater possibility of a communication–action

disconnect.

GI is consequently more closely aligned with the action side of the action–performance construct than raw disclosure counts, word frequency, sentiment, or overall environmental disclosure volume. A recent systematic review identifies measures based on disclosure quantity, report length, sustainability-word frequency, and sentiment as common ways to proxy for firms' symbolic environmental communication (Lublóy et al., 2025). For example, the widely used decoupling measure of Yu et al. (2020) contrasts an ESG disclosure score—which reflects the amount of disclosed ESG data—with an ESG-performance score. Content-analysis studies similarly use report length, the frequency or proportion of sustainability terms, and sentiment to capture the extent, specificity, or tone of corporate communication (Grewal et al., 2022; Walker and Wan, 2012; Baier et al., 2020; Bilinski, 2024). These measures are valuable for capturing reporting intensity or rhetorical positivity, but quantity- and tone-based measures do not, on their own, distinguish firm-specific implementation from symbolic expression. A report may discuss a sustainability topic extensively, or use many favourable environmental terms, while masking other material environmental issues or providing no evidence that the firm implemented a project, adopted a technology, remediated pollution, or achieved a specified environmental outcome (Lublóy et al., 2025; Clarkson et al., 2008; Hummel and Schlick, 2016; Xu et al., 2026). By semantically classifying environmental statements as substantive or symbolic, GI retains the scale of classified communication in its denominator while identifying the action-oriented component in its numerator. It therefore separates disclosure quantity from the substantive content of disclosure, which is the particular communication dimension relevant to our action–performance measure.

We identify greenwashing as the within-industry-year divergence between GI and an external environmental-performance score: greenwashing only when a relatively high GI is not matched by comparatively strong environmental performance (Delmas and Burbano, 2011; Lyon and Maxwell, 2011; Lublóy et al., 2025; Xu et al., 2026).

2.2 Environmental Action Claims as Points of Comparison

The hard–soft disclosure distinction gives the measurement argument a direct auditing implication. Hard environmental information refers to relatively specific and verifiable content, such as environmental investments, performance indicators, expenditures, incidents, or progress toward specified targets. Soft information consists more often of general policies, principles, commitments, and favorable descriptions (Clarkson et al., 2008; Hummel and Schlick, 2016). The distinction matters here not simply because hard information is substantively richer, but because its specificity creates identifiable objects that an auditor may compare with financial statements and with knowledge obtained during the audit.

For example, a claim that an environmental equipment project has been implemented may have implications for capital expenditures, construction in progress, procurement, or payments. A pollution-remediation claim may relate to expenditures, environmental liabilities, provisions, or contingencies (Barth and McNichols, 1994; Barth et al., 1997; Li et al., 1997), while an environmental technology upgrade may be connected to equipment changes, production processes, operating costs, or industry-specific technical conditions (Clarkson et al., 2004). Symbolic statements generally lack these identifiable implications. Thus, GI links the measurement to the audit mechanism because it identifies the portion of environmental communication that is more likely to generate a concrete comparison point with information available to the auditor.

CAS 1521 establishes the relevant procedural boundary. When reading other information, the audi-

tor considers whether it is materially inconsistent with the financial statements or with knowledge obtained during the audit and remains alert to indications that the other information may be materially misstated. If an apparent material inconsistency arises, the auditor discusses the matter with management and considers whether additional procedures or communication are necessary. Accordingly, CAS 1521 does not require auditors to verify all environmental claims or to provide sustainability assurance. It does, however, provide a limited channel through which a concrete environmental claim may become relevant to the auditor when that claim intersects with available financial, operational, or compliance-related information.

2.3 Industry Specialization and Recognition of Inconsistencies

Industry specialization strengthens the information base relevant to this potential comparison. Through repeated engagements, specialist audit firms develop knowledge about industry business models, production processes, regulatory requirements, technologies, cost structures, and reporting risks (Francis, 2011; Reichelt and Wang, 2010). This knowledge helps auditors form more precise expectations about the relationships among a client's operations, financial information, and industry conditions. Experimental evidence indicates that industry specialists possess richer industry knowledge and are better able to identify errors or unusual patterns in industry-related information (Solomon et al., 1999; Owhoso et al., 2002). Industry specialization also affects audit risk assessments and planning decisions, including judgments about the nature and quality of subsequent audit procedures (Low, 2004). Hammersley (2006) further shows that industry specialists can identify complex error patterns formed by several individually ambiguous cues and can propose more discriminating follow-up procedures (Hammersley, 2006).

These studies support a inference about comparative recognition ability. When an environmental action claim, a financial reporting item, and an operational or regulatory fact jointly form an unusual pattern, industry specialists may be better positioned to recognize that the pattern warrants attention. For example, an assertion that a major environmental upgrade has been completed may be more informative to an industry specialist who understands the technology, production scale, expected investment, and operational consequences associated with that upgrade. Similarly, a claim of comprehensive environmental rectification may be easier to evaluate when the auditor understands the client's exposure to industry regulation and the operational implications of the alleged rectification. The relevant mechanism is that concrete claims increase the opportunity for comparison, while industry specialization improves the auditor's ability to interpret whether the available information forms a potentially inconsistent pattern.

2.4 From Recognition to Informed Questioning

Recognition of a potential inconsistency may affect the information environment surrounding management's disclosure choices. Under CAS 1521, an auditor who identifies an apparent material inconsistency discusses the matter with management, considers the explanations and supporting information provided, and determines whether further action or communication is necessary. Industry knowledge can make these questions more specific because the auditor can relate management's explanation to the client's production activities, financial reporting, technology, and regulatory setting. More informed questioning implies that management may face greater difficulty sustaining a concrete claim that is difficult to reconcile with information already available to the auditor.

This reasoning is consistent with evidence that auditing can influence disclosure choices beyond the audited financial statements and that auditor industry specialization is associated with client dis-

closure quality (Schroeder, 2016; Dunn and Mayhew, 2004). A firm may still disclose environmental actions for legitimacy or capital-market reasons, but an industry specialist may increase the expected cost of presenting unsupported action claims as evidence of strong environmental performance. The resulting prediction is an association between specialization and less greenwashing.

2.5 Scope Conditions and Hypothesis

Several considerations limit this predicted association. The auditor's responsibility for other information is narrower than the responsibility arising in a separate sustainability assurance engagement. Financial-statement audit expertise may also be insufficient to evaluate the scientific or technical substance of some environmental claims. Some environmental actions may be implemented through arrangements that have no clear or material accounting manifestation. Firms also retain strong legitimacy and capital-market incentives to maintain favorable environmental narratives (Delmas and Burbano, 2011; Healy and Palepu, 2001; Verrecchia, 2001). Dedicated sustainability assurance remains a distinct governance mechanism with different procedures, expertise, and responsibilities (Simnett et al., 2009; Pflugrath et al., 2011; Cohen and Simnett, 2015). These limitations imply that industry specialization should not be expected to eliminate greenwashing or to affect every environmental statement. They do not, however, rule out a detectable average association when firms' concrete environmental claims provide points of comparison and specialist auditors possess a comparative advantage in recognizing inconsistent patterns. We therefore state the following hypothesis:

H1: Audit firm industry specialization is negatively associated with corporate greenwashing.

3 Research Design

3.1 Sample Selection and Data Sources

Our initial sample consists of Chinese A-share listed firms from 2021 through 2024. We exclude firms designated as *ST, ST, or PT during the sample period, as these firms may exhibit abnormal financial conditions and operating characteristics. Annual reports are obtained from the CNINFO website, and financial, corporate-governance, audit, and full-market audit-client data are primarily obtained from the China Stock Market & Accounting Research (CSMAR) database. We use Huazheng environmental scores as the external measure of firms' environmental performance. The annual-report text and environmental scores are used to construct the greenwashing measure, while the full-market audit-client data are used to calculate audit firm industry specialization. Financial, governance, and audit data provide the control variables. Following the industry-classification convention commonly applied to Chinese listed firms (Gul et al., 2010), we classify manufacturing firms at the CSRC two-digit industry level and firms in other sectors at the CSRC one-digit industry level.

Our sample-selection procedure is designed to ensure the completeness of the annual-report text, the feasibility of the green-disclosure classification, and the availability of data required for the empirical analysis. We first retain firm-years for which the relevant annual-report text can be extracted completely and the green-disclosure identification procedure can be completed. Firm-years with extraction or classification failures are treated as incomplete rather than as having zero green disclosure. We combine the valid textual classifications with Huazheng environmental scores to

form the dependent-variable panel. We then match this panel to audit firm industry-specialization measures calculated from the full A-share audit market. Finally, we exclude observations with substantial missing financial, governance, or audit data and require nonmissing values for the dependent variable, both main specialization measures, all control variables, firm and year identifiers, and the industry code used in the analysis. Table 1 summarizes the sample-selection procedure. Following common practice, we winsorize continuous control variables at the 1st and 99th percentiles to limit the influence of extreme accounting ratios and growth rates. We do not winsorize the specialization measures because their upper-tail values represent economically meaningful audit-market concentration rather than mechanical outliers. We also retain the original Greenwashing measure because its tail observations capture firms with unusually large communication–performance gaps, which constitute the substantive phenomenon examined in this study.

Table 1: Sample Selection

Sample-selection procedure	Firm-years
Initial A-share firm-year panel, 2021–2024	18,077
Less: incomplete annual-report text extraction or unsuccessful green-disclosure classification, or insufficient industry-year variation to construct the continuous greenwashing measure	(1,594)
Less: missing financial control variables	(700)
Less: missing corporate-governance control variables	(381)
Less: missing audit control variables	(21)
Final regression sample	15,381

Notes: The initial panel excludes firms designated as `*ST`, `ST`, or `PT` during the sample period. Firm-years with incomplete text extraction or unsuccessful green-disclosure classification are treated as incomplete observations rather than as zero environmental disclosure. The final sample contains 4,604 unique firms over 2021–2024.

3.2 Variable Measurement

3.2.1 Measure of Greenwashing

We construct the dependent variable by adapting the DeepGreen procedure (Xu et al., 2026). We first extract the "Environmental Information" section from each annual report[1]. Layer A identifies environmentally relevant terms, after which we retrieve the complete sentence containing each identified term from the unsegmented report text. Layer B classifies each resulting keyword–sentence pair as substantive or symbolic environmental disclosure [2]. The reliability of the semantic classifications is assessed through an independent human-validation exercise reported in Appendix A.

Substantive disclosure describes a concrete environmental action undertaken by the firm or its subsidiaries, such as implementing a project, adopting a technology, or taking a specific pollution-control measure. Symbolic disclosure includes general aspirations, slogans, policy restatements, and other statements that do not identify a concrete corporate action. Conventional measures based on the total number of environmental words or statements capture disclosure quantity but cannot distinguish substantive environmental actions from symbolic language. We therefore identify environmental statements and classify them according to whether they describe a concrete firm-level action. Let the total number of classified environmental statements be the sum of substantive

[1]We segment the Chinese text using Jieba, remove terms contained in the HIT Stopword List

[2]The two LLM classification layers using Qwen 3.6, a Qwen3-family model (Yang et al., 2025).

statements and symbolic statements. The total reflects the scale of environmental communication, while the substantive share—GI—identifies the effective portion of that communication. When GI approaches one, the disclosure is predominantly action-oriented; when GI approaches zero, it is predominantly symbolic or absent. This makes GI particularly suitable for capturing the disclosure-side effectiveness of an action–performance divergence, while the external Huazheng score supplies the separate performance dimension.

We aggregate the Layer B classifications at the firm-year level. Let $S_{i,t}$ denote the number of keyword–sentence pairs classified as substantive and $P_{i,t}$ the number classified as symbolic. The green implementation ratio is

$$GI_{i,t} = \frac{S_{i,t}}{S_{i,t} + P_{i,t}}.$$

The unit of analysis is the keyword–sentence pair rather than the unique sentence; consequently, a sentence containing multiple environmental keywords may contribute more than one classification. Observations for which the text-extraction or classification procedure does not yield any valid keyword–sentence pairs are treated as missing rather than as zero environmental disclosure.

Our primary measure compares environmental communication with external environmental performance. Firms in different industries may have systematically different opportunities, technologies, regulatory exposure, and disclosure practices; standardizing across the pooled sample could therefore confound relative greenwashing with cross-industry differences. Following the recommendation that communication and performance components be benchmarked against industry peers rather than the pooled sample (Lublóy et al., 2025; Yu et al., 2020), we standardize GI and the Huazheng environmental score within each industry-year group. The industry classification is the same as that used for audit firm industry specialization. This within-industry-year normalization follows the peer-relative logic of prior decoupling measures, while addressing the concern that disclosure practices and environmental performance differ systematically across industries (Yu et al., 2020; Lublóy et al., 2025). For each component, the z-score is calculated using the corresponding industry-year mean and sample standard deviation ($ddof = 1$). We define

$$Greenwashing_{i,t} = GI_z_{i,t} - ESGE_z_{i,t}.$$

This difference measure follows the broader decoupling approach that compares a communication-side indicator with an independently assessed performance indicator (Walker and Wan, 2012; Yu et al., 2020; Lublóy et al., 2025). A higher value indicates that a firm's substantive share of green communication is high relative to its industry-year peers while its externally assessed environmental performance is comparatively low, and therefore represents a larger communication–performance gap. As an alternative measure, we construct a binary indicator equal to one when GI is strictly above its industry-year mean and the Huazheng environmental score is strictly below its industry-year mean, and zero otherwise(Xu et al., 2026).

3.2.2 Measures of Audit Firm Industry Specialization

Our explanatory variable measures specialization at the audit-firm level. Audit firms are identified by the stable CSMAR institution identifier. Following the market-share approach used in prior industry-expertise research (Cai and Xian, 2007; Krishnan, 2003), we calculate specialization from the full A-share audit market.

Let $Revenue_{i,t}$ be client operating revenue, a index audit firms, and j index industries. Our primary continuous measure, IMS, is

$$IMS_{a,j,t} = \frac{\sum_{i\in(a,j,t)} Revenue_{i,t}}{\sum_{i\in(j,t)} Revenue_{i,t}}.$$

The numerator is the operating revenue of all valid A-share clients audited by firm a in industry j and year t, and the denominator is the operating revenue of all valid A-share firms in that industry-year. We include only observations with positive operating revenue and nonmissing audit-firm and industry identifiers. The resulting audit-firm–industry–year measure is matched to each client using year, audit-firm identifier, and this mixed industry code. An unmatched specialization observation remains missing.

We define IMS_D as one when IMS is at least 20 percent and zero otherwise.

3.2.3 Control Variables

We include financial, governance, and audit controls that may be associated with firms' environmental disclosure and performance or with the selection of specialist auditors (Marquis et al., 2016; Reichelt and Wang, 2010; DeFond and Zhang, 2014). Financial controls comprise firm size ($Size$), measured as the natural logarithm of total assets; leverage (Lev), measured as total liabilities divided by total assets; return on assets (ROA), measured as net profit divided by total assets; operating cash flow (CFO), measured as operating net cash flow divided by total assets; sales growth ($Growth$), measured as current-year operating revenue divided by prior-year operating revenue minus one; and listing age ($FirmAge$), measured as the number of years since the firm's initial public listing. Growth is first calculated using consecutive-year revenue observations in the full-market financial data and is then matched to the research panel, which avoids creating artificial lags within the selected sample.

Governance controls comprise state ownership (SOE), an indicator equal to one when the CSMAR actual-controller type is state-owned and zero otherwise; CEO–chair duality ($Dual$); board size ($BoardSize$), measured as the natural logarithm of the number of directors; and the proportion of independent directors ($Indep$). Audit controls comprise an indicator for a Big Four auditor ($Big4$); audit fees ($LnAuditFee$), measured as the natural logarithm of total audit fees; auditor tenure ($AuditorTenure$); and an indicator for an audit-firm change ($IsChangeFirm$), coded as one if the firm appoints a different audit firm in the current year than in the preceding year, and zero otherwise. These audit controls help distinguish industry specialization from general audit-firm brand, audit effort, accumulated client knowledge, and changes in the auditor–client relationship (DeFond and Zhang, 2014). Values that do not satisfy the required positive-denominator or logarithm conditions, observations with nonconsecutive years for growth, and undefined or infinite ratios are treated as missing.

3.3 Empirical Model

We estimate the following pooled OLS model:

$$Greenwashing_{i,t} = \alpha + \beta Specialization_{i,t} + \gamma' Controls_{i,t} + YearFE_t + IndustryFE_j + \varepsilon_{i,t}.$$

The dependent variable is the continuous greenwashing measure. $Specialization$ is alternatively IMS and IMS_D, and $Controls$ contains the financial, governance, and audit variables defined

above. The primary specification includes year and industry fixed effects, using the industry classification described above, and standard errors clustered by firm. H1 predicts $\beta < 0$. We estimate successive specifications containing only specialization, then adding financial controls, governance controls, audit controls, and finally year and industry fixed effects. We also estimate the complete model with industry-clustered standard errors. All specifications use the same 15,381 firm-year observations.

4 Empirical Results

4.1 Descriptive Statistics

Table 2 presents descriptive statistics for the common regression sample. Greenwashing has a mean close to zero by construction because both its components are standardized within industry-year groups. The measure captures the relative distance between environmental communication and environmental performance. Its standard deviation is 1.337, and the interquartile range extends from -0.820 to 0.898. This dispersion indicates considerable variation across firms in the degree to which their substantive environmental communication exceeds or falls short of their relative environmental performance. The mean IMS is 0.070, while 5.7 percent of firm-years are audited by firms meeting the 20 percent strong-specialization threshold. This tail share indicates identifiable variation in strong industry expertise and provides a basis for comparing clients of strong specialists with other clients. The average leverage ratio is 40.97 percent, suggesting that firms face debt-related financial pressure. State-controlled firms account for 27.7 percent of the sample, highlighting the importance of regulatory and political expectations in shaping environmental disclosure. Big Four clients account for only 5.5 percent of firm-years, whereas auditor changes occur in 11.5 percent, suggesting that most firms are served by non-Big Four auditors and maintain relatively stable auditor–client relationships.

4.2 Correlation Analysis

Table 3 reports the pairwise correlations. Greenwashing is negatively correlated with IMS (Pearson = -0.054) and IMS_D (Spearman = -0.033), providing preliminary evidence consistent with H1: clients of auditors with greater industry expertise tend to exhibit a smaller gap between environmental communication and environmental performance. These unconditional associations are modest, as expected because firms' financial characteristics, governance, auditor choice, and industry conditions may jointly affect both specialization and greenwashing; the multivariate analysis therefore evaluates whether the association persists after accounting for these factors. The 0.402 correlation between IMS and IMS_D reflects that IMS_D is a threshold-based representation of IMS. Among the controls, the largest correlation is 0.698 between Size and LnAuditFee, consistent with larger firms requiring greater audit effort and paying higher audit fees.

4.3 Baseline Regression Results

We estimate a sequence of nested regression models to examine the association between audit firm industry specialization and corporate greenwashing. Model (1) includes only the specialization measure, IMS or IMS_D, and provides the unconditional association between auditor industry specialization and greenwashing. Model (2) adds financial controls, including firm size, leverage, profitability, operating cash flow, sales growth, and firm age, to account for differences in firms' financial condition, operating performance, and life cycle. Model (3) further incorporates corporate-

governance controls, including state ownership, CEO–chair duality, board size, and the proportion of independent directors, because governance structures may affect both auditor selection and the credibility of environmental disclosure. Model (4) adds audit-related controls, including Big Four affiliation, audit fees, auditor tenure, and auditor changes, allowing the analysis to distinguish industry specialization from auditor reputation, audit effort, accumulated client knowledge, and changes in the auditor–client relationship. All four model specifications include year and industry fixed effects.

Tables 4 and 5 report the baseline regression results. Across all specifications, the coefficients on both IMS and IMS_D are negative, consistent with H1. In the most comprehensive specification, column (4), the coefficient on IMS is -0.4829 and statistically significant at the 5 percent level. As reported in Table 2, IMS is right-skewed, with a 25th percentile of 0.015 and a 75th percentile of 0.098. Moving from the 25th to the 75th percentile—an increase of 0.083 in the auditor's share of industry operating revenue—is associated with a 0.040 decline in Greenwashing, or about 0.03 of the outcome's standard deviation (1.337).

This interquartile increase is nearly identical to a one-standard-deviation increase of 0.082, which corresponds to a 0.039 decline. The coefficient on IMS_D in column (4) is -0.1434 and significant at the 10 percent level. Thus, relative to other clients, firms audited by auditors meeting the 20 percent threshold have Greenwashing scores that are 0.143 lower, or about 0.11 of a standard deviation. In the same specification, the Big4 coefficient is -0.3613, or about 0.27 of a standard deviation. This larger association is consistent with the reputation capital of brand-name auditors, which raises the expected cost of leaving unsupported environmental claims in other information unchallenged (DeFond and Zhang, 2014). A one-standard-deviation increase in Size (1.215) is associated with a 0.165 decline in Greenwashing, or about 0.12 of a standard deviation. This association is consistent with the greater visibility and external scrutiny of larger firms, which make a communication–performance gap more costly to sustain (Marquis et al., 2016). The negative and statistically significant coefficients across the two alternative specialization measures provide support for H1. As an additional check, Appendix B examines whether the baseline inference is sensitive to alternative standard-error clustering schemes.

Table 2: Descriptive Statistics

Variable	N	Mean	Std. Dev.	Min	25th Pctl.	Median	75th Pctl.	Max
Greenwashing	15,381	-0.0021	1.3366	-7.4995	-0.8204	-0.0291	0.8979	5.0962
IMS	15,381	0.0698	0.0817	0.0000	0.0152	0.0420	0.0983	0.7307
IMS_D	15,381	0.0573	0.2324	0.0000	0.0000	0.0000	0.0000	1.0000
Size	15,381	22.3249	1.2148	20.1877	21.4376	22.1117	23.0181	26.0914
Lev	15,381	0.4097	0.2015	0.0557	0.2490	0.4012	0.5534	0.8994
ROA	15,381	0.0269	0.0644	-0.2226	0.0064	0.0310	0.0608	0.1909
CFO	15,381	0.0477	0.0644	-0.1363	0.0110	0.0459	0.0847	0.2311
Growth	15,381	0.0957	0.3074	-0.5521	-0.0708	0.0605	0.2086	1.5243
FirmAge	15,381	11.2589	8.7381	0.0000	4.0000	9.0000	17.0000	30.2000
SOE	15,381	0.2768	0.4474	0.0000	0.0000	0.0000	1.0000	1.0000
Dual	15,381	0.3439	0.4750	0.0000	0.0000	0.0000	1.0000	1.0000
BoardSize	15,381	2.1129	0.2198	1.6094	1.9459	2.1972	2.1972	2.7081
Indep	15,381	0.3826	0.0589	0.2727	0.3333	0.3750	0.4286	0.5714
Big4	15,381	0.0555	0.2289	0.0000	0.0000	0.0000	0.0000	1.0000
AuditFee	15,381	14.0021	0.5682	12.8992	13.5924	13.9108	14.3163	15.8304
Tenure	15,381	7.9131	6.0436	1.0000	3.0000	6.0000	11.0000	27.0000
AuditorChange	15,381	0.1151	0.3191	0.0000	0.0000	0.0000	0.0000	1.0000

Notes: This table reports descriptive statistics for all variables. All statistics are based on 15,381 firm-year observations.

Table 3: Pairwise Correlations

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)
(1) Greenwashing	1.000																
(2) IMS	**-0.054**	1.000															
(3) IMS_D	**-0.033**	**0.402**	1.000														
(4) Size	**-0.113**	**0.163**	**0.105**	1.000													
(5) Lev	-0.013	-0.003	**0.024**	**0.430**	1.000												
(6) ROA	**-0.017**	**0.099**	**0.036**	**0.106**	**-0.347**	1.000											
(7) CFO	**-0.023**	**0.101**	**0.064**	**0.124**	**-0.153**	**0.489**	1.000										
(8) Growth	0.000	0.001	-0.013	**0.033**	**0.034**	**0.300**	**0.082**	1.000									
(9) FirmAge	**-0.025**	**-0.021**	**0.039**	**0.429**	**0.310**	**-0.125**	**-0.018**	**-0.074**	1.000								
(10) SOE	**-0.034**	**-0.029**	**0.029**	**0.337**	**0.231**	**-0.072**	-0.014	**-0.044**	**0.443**	1.000							
(11) Dual	0.001	**0.022**	**-0.031**	**-0.190**	**-0.118**	0.011	-0.007	**0.027**	**-0.250**	**-0.310**	1.000						
(12) BoardSize	**-0.019**	**0.036**	**0.036**	**0.276**	**0.131**	**0.035**	**0.050**	0.009	**0.218**	**0.306**	**-0.198**	1.000					
(13) Indep	-0.010	0.004	-0.004	**-0.074**	**-0.046**	-0.001	0.002	-0.001	**-0.067**	**-0.140**	**0.139**	**-0.430**	1.000				
(14) Big4	**-0.080**	**0.118**	**0.120**	**0.211**	**0.059**	**0.044**	**0.085**	0.009	**0.040**	**0.070**	**-0.037**	**0.074**	-0.013	1.000			
(15) AuditFee	**-0.100**	**0.168**	**0.112**	**0.698**	**0.382**	**-0.055**	**0.087**	**-0.025**	**0.313**	**0.164**	**-0.098**	**0.178**	**-0.033**	**0.299**	1.000		
(16) Tenure	**0.034**	**0.081**	**0.051**	**0.069**	0.011	**0.062**	**0.061**	-0.006	**0.135**	**-0.184**	**0.041**	**0.028**	**-0.027**	**-0.060**	**0.068**	1.000	
(17) AuditorChange	-0.011	**-0.068**	0.007	**0.059**	**0.053**	**-0.071**	**-0.034**	**-0.044**	**0.122**	**0.170**	**-0.057**	**0.052**	**-0.029**	**0.022**	**0.030**	**-0.554**	1.000

Notes: This table reports Pearson correlations when both variables are continuous and Spearman correlations when either variable is binary. Coefficients in bold are significant at the 5 percent level using two-tailed tests. Variables are numbered as follows: (1) Greenwashing, (2) IMS, (3) IMS_D, (4) Size, (5) Lev, (6) ROA, (7) CFO, (8) Growth, (9) FirmAge, (10) SOE, (11) Dual, (12) BoardSize, (13) IndDirector, (14) Big4, (15) AuditFee, (16) Tenure, and (17) AuditorChange. All statistics are based on 15,381 firm-year observations.

Table 4: Audit Firm Industry Specialization and Corporate Greenwashing: IMS

	(1)	(2)	(3)	(4)
IMS	-1.0488***	-0.5396**	-0.5433**	-0.4829**
	(0.2334)	(0.2348)	(0.2351)	(0.2362)
Size		-0.1708***	-0.1690***	-0.1362***
		(0.0207)	(0.0211)	(0.0246)
Lev		0.3033***	0.3087***	0.2931***
		(0.1082)	(0.1085)	(0.1092)
ROA		0.5571**	0.5666**	0.4111
		(0.2794)	(0.2792)	(0.2796)
CFO		-0.2045	-0.2217	-0.1424
		(0.2298)	(0.2294)	(0.2301)
Growth		-0.0002	-0.0020	0.0019
		(0.0394)	(0.0394)	(0.0392)
FirmAge		0.0017	0.0030	0.0007
		(0.0025)	(0.0027)	(0.0028)
SOE			-0.0891*	-0.0722
			(0.0508)	(0.0515)
Dual			-0.0321	-0.0327
			(0.0364)	(0.0363)
BoardSize			0.0318	0.0362
			(0.0809)	(0.0803)
Indep			-0.3352	-0.2725
			(0.2878)	(0.2871)
Big4				-0.3613***
				(0.0960)
AuditFee				-0.0441
				(0.0467)
Tenure				0.0099***
				(0.0033)
AuditorChange				0.0562
				(0.0392)
Intercept	0.0602	3.6559***	3.7020***	3.5218***
	(0.1769)	(0.4665)	(0.5056)	(0.6184)
Observations	15,381	15,381	15,381	15,381
Adjusted R^2	0.0009	0.0157	0.0163	0.0220
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes

Notes: The dependent variable is continuous Greenwashing. All specifications include year and industry fixed effects and use firm-clustered standard errors. Coefficients are reported above standard errors in parentheses. ***, **, and * indicate two-tailed significance at 1, 5, and 10 percent.

Table 5: Audit Firm Industry Specialization and Corporate Greenwashing: IMS_D

	(1)	(2)	(3)	(4)
IMS_D	-0.2942***	-0.1711**	-0.1710**	-0.1434*
	(0.0825)	(0.0795)	(0.0795)	(0.0794)
Size		-0.1747***	-0.1730***	-0.1390***
		(0.0204)	(0.0208)	(0.0245)
Lev		0.3079***	0.3133***	0.2982***
		(0.1082)	(0.1085)	(0.1091)
ROA		0.5444*	0.5534**	0.3967
		(0.2791)	(0.2790)	(0.2795)
CFO		-0.2193	-0.2362	-0.1534
		(0.2302)	(0.2298)	(0.2306)
Growth		0.0002	-0.0016	0.0021
		(0.0395)	(0.0394)	(0.0392)
FirmAge		0.0021	0.0033	0.0010
		(0.0025)	(0.0027)	(0.0028)
SOE			-0.0882*	-0.0724
			(0.0509)	(0.0515)
Dual			-0.0332	-0.0335
			(0.0364)	(0.0363)
BoardSize			0.0291	0.0339
			(0.0809)	(0.0803)
Indep			-0.3342	-0.2719
			(0.2879)	(0.2872)
Big4				-0.3604***
				(0.0960)
AuditFee				-0.0472
				(0.0467)
Tenure				0.0098***
				(0.0032)
AuditorChange				0.0584
				(0.0392)
Intercept	0.0170	3.7177***	3.7701***	3.6083***
	(0.1745)	(0.4632)	(0.5025)	(0.6146)
Observations	15,381	15,381	15,381	15,381
Adjusted R^2	-0.0004	0.0156	0.0161	0.0219
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes

Notes: The dependent variable is continuous Greenwashing. All specifications include year and industry fixed effects and use firm-clustered standard errors. Coefficients are reported above standard errors in parentheses. ***, **, and * indicate two-tailed significance at 1, 5, and 10 percent.

5 Additional Analyses Addressing Endogeneity

Although the baseline models include an extensive set of financial, governance, and audit controls, together with year and industry fixed effects, client–auditor matching, reverse causality, and unobserved factors may still affect the estimates. The following analyses are designed to improve temporal ordering and mitigate observable-selection bias; they do not eliminate endogeneity. We therefore interpret the results as supplementary evidence on the association between audit firm industry specialization and greenwashing.

5.1 Lagged Covariates and Temporal Ordering

Firm characteristics, governance arrangements, disclosure practices, and auditor–client relationships are dynamic rather than predetermined constants. They may change in response to operating performance, regulatory scrutiny, financing needs, or the information environment, while the same forces may also be associated with greenwashing and the choice of auditor specialization. Treating all of these variables as contemporaneous controls can therefore create ambiguity about whether a control is a prior determinant of the outcome, a simultaneous correlate, or partly a response to the same conditions that shape current Greenwashing. More generally, temporal ordering is important in observational panel research because persistence in firm policies and reporting outcomes can make current associations reflect both current decisions and previously established firm characteristics. A lagged-covariate specification provides a more demanding test of whether the association remains after conditioning on observable characteristics that were measured before the outcome year (Bates, 2012).

We therefore re-estimate the baseline model using the preceding fiscal year's covariates. The procedure follows exactly the baseline operation: we first retain the 15,381 firm-years in the baseline common sample and winsorize continuous control variables at the 1st and 99th percentiles within that sample, and then construct lagged variables only when adjacent firm-year observations are strictly consecutive. The resulting sample contains 10,488 firm-years from 4,168 firms, spanning three years and 47 industries. Following the baseline regression, the specification includes all three classes of controls—financial, governance, and audit variables—together with year and industry fixed effects and firm-clustered standard errors.

The dependent variable is current Greenwashing, and current IMS is retained as the focal explanatory variable. We retain current IMS for two reasons. First, the audit-firm specialization measure describes the auditor serving the client in the outcome year, so matching the treatment year to the Greenwashing year preserves the interpretation of the baseline association. Second, the purpose of this test is to move the observable covariates, rather than the focal specialization measure, to the period preceding the outcome while otherwise holding the baseline specification as constant as possible. The model uses the preceding-year values of Size, Lev, ROA, CFO, Growth, FirmAge, SOE, Dual, BoardSize, IndependentDirectorRatio, Big4, LnAuditFee, AuditorTenure, and IsChangeFirm.

As reported in Model (1) of Table 7, the coefficient on IMS is -0.5162 (standard error = 0.2515; $t = -2.0523$; $p = 0.0401$). The adjusted R^2 is 0.0244. Thus, the negative association remains after prior-year observable financial, governance, and audit characteristics are taken into account. This result improves the temporal ordering of the covariates and reduces the concern that current Greenwashing mechanically determines the measured control variables.

5.2 Generalized Propensity-Score Weighting for Continuous Specialization

The baseline regression conditions on observed firm, governance, and audit characteristics, but clients may still sort into auditors with different levels of industry specialization according to their disclosure histories, operating risks, governance structures, audit demands, or existing auditor relationships. Propensity-score methods address this concern by modeling treatment assignment from observed pre-treatment characteristics and reweighting the sample so that firms with different treatment intensities are more comparable on those characteristics (Shipman et al., 2017).

Because IMS is a continuous audit-market share, reducing it to a treated-versus-untreated indicator would discard differences in specialization intensity. We therefore use the generalized propensity score (GPS), defined as the conditional density of a continuous treatment given pre-treatment covariates (Imbens, 2000; Hirano and Imbens, 2004). An observation receives a larger weight when its realized specialization is less likely given its preceding-year characteristics, and the weights are stabilized by the marginal density of the transformed treatment (Robins et al., 2000). The same approach has been used to address selection on observables when the treatment is continuous (Campello et al., 2014; Brick et al., 2024). In this setting, the GPS–IPW weights adjust the client–auditor comparison for observable selection.

The first-stage covariates are lagged Greenwashing and the lagged financial, governance, and audit-selection variables from the baseline specification, together with year and industry fixed effects. Because IMS lies between zero and one, the treatment model is estimated on its logit transformation. We require strictly consecutive firm-year observations when constructing lagged variables, so that a nominally preceding row is not treated as $t-1$ when an intervening year is missing. This procedure yields 10,488 firm-years from 4,168 firms, fewer than the 15,381 observations in the baseline sample because the GPS panel requires valid pre-treatment information and strict temporal continuity. Appendix C details the first-stage model, the construction of the stabilized weights, and the effective sample size.

Before estimating the weighted outcome regression, we assess whether the GPS weights improve comparability across specialization intensities. For this diagnostic only, we divide IMS into quintile dose groups and compare each group with the remaining observations using standardized mean differences (SMDs). Table 6 reports, for every first-stage covariate, the mean and maximum absolute SMD across the five dose groups before and after weighting. The weighted mean absolute SMD is lower for 11 of the 15 covariates, and the maximum absolute SMD is also lower for 12 of the 15 covariates. These diagnostics indicate that the GPS weights improve observable comparability.

After assessing balance, we re-estimate the baseline model using the GPS weights, with standard errors clustered by firm. Consistent with the staged reporting logic of the baseline regressions, Table 7 reports this weighted model in two steps. Model (2) includes the specialization measure, GPS weights, and year and industry fixed effects but omits the outcome-equation controls. Model (3) adds the complete set of current financial, governance, and audit controls. The IMS coefficient is -0.8549 in Model (2) (standard error $= 0.2737$; $p = 0.0018$) and -0.2878 in Model (3) (standard error $= 0.1191$; $p = 0.0157$). Both models use 10,488 firm-years from 4,168 firms, and the untrimmed effective sample size is 7,485. The coefficient remains negative after the complete set of current controls is introduced, consistent with the baseline association after an observable-selection adjustment. Appendix C also reports the P0.5–P99.5 and P1–P99 weight-trimming results, together with binary entropy-balancing and propensity-score matching complements that treat specialization as IMS_D at the 20 percent threshold.

This analysis mitigates observable-selection bias. Residual imbalance at the distributional extremes, unobserved client–auditor matching factors, unmeasured auditor information, reverse causality, and common industry-year shocks may remain, and the GPS weights do not remove endogeneity from unobservable covariates (Brick et al., 2024). The GPS results are therefore supplementary evidence consistent with a negative association.

5.3 Auditor-Switching Evidence

Auditor changes reflect a client–auditor matching decision rather than an exogenously assigned event (DeFond and Subramanyam, 1998). We therefore examine verified switching events, requiring both $IsChangeFirm = 1$ [3] and an observed change in $InstitutionID$ [4] between adjacent years. The input panel contains 15,381 observations and 4,604 firms. Following the baseline procedure, continuous controls are winsorized at the 1st and 99th percentiles using the complete baseline sample before lag construction. After verification, the principal switching sample contains 1,337 observations from 1,247 firms, including 98 switches into specialist auditors. The defined $UpgradeExpert$ indicator, which identifies moves from a non-specialist to a specialist auditor, comprises 69 firm-year events.

We first evaluate whether a firm's pre-switch Greenwashing is systematically associated with the decision to switch into a specialist. Let $SwitchInExpert_{i,t}$ equal one when a verified switching firm appoints a specialist auditor in year t, and zero when it appoints a non-specialist auditor. Our selection equation is the Logit model,

$$\Pr(SwitchInExpert_{i,t} = 1) = \Lambda\left(\alpha + \beta L_Greenwashing_{i,t} + \theta' X_{i,t-1} + \lambda_t + \delta_j\right),$$

where $\Lambda(\cdot)$ is the logistic cumulative distribution function, $X_{i,t-1}$ contains the same full set of financial, governance, and audit controls used in the baseline regression, each measured at $t-1$, and $L_Greenwashing_{i,t}$ is the lagged Greenwashing measure. The selection equation accounts for the possibility that a firm's pre-switch reporting environment and observable characteristics may influence auditor choice (DeFond and Subramanyam, 1998). Because the selection outcome records whether the firm switches into a specialist, it is a binary auditor-choice indicator; the Logit specification models is discrete selection outcome and constrains fitted probabilities to the unit interval (Wooldridge, 2010). The fixed-effects and standard-error treatment follows the baseline regression design. As reported in Table 8, the estimated coefficient on lagged Greenwashing is 0.0852 ($p = 0.3547$; $N = 777$), providing no evidence that pre-switch Greenwashing systematically predicts switching into a specialist.

The selection equation addresses aspects of observable sorting into specialist auditors. It does not eliminate unobserved client–auditor matching factors, reverse causality, or concurrent factors associated with switching.

[3] $IsChangeFirm$ equals one when the firm appoints a different audit firm in the current year than in the preceding year, and zero otherwise.

[4] $InstitutionID$ is the stable audit-institution identifier used to verify whether those records refer to different audit firms.

Table 6: GPS Covariate Balance Before and After Weighting

Covariate	Mean $\lvert SMD \rvert$ Before	Mean $\lvert SMD \rvert$ After	Max $\lvert SMD \rvert$ Before	Max $\lvert SMD \rvert$ After
L_Greenwashing	0.0447	0.0270	0.0923	0.0582
L_Size	0.1628	0.0747	0.3597	0.1670
L_Lev	0.0342	0.0488	0.0703	0.1042
L_ROA	0.1419	0.0376	0.2937	0.0717
L_CFO	0.1123	0.0312	0.1913	0.0490
L_Growth	0.0561	0.0630	0.0841	0.1048
L_FirmAge	0.0770	0.0382	0.1401	0.0523
L_SOE	0.0615	0.0592	0.1269	0.0904
L_Dual	0.0362	0.0264	0.0644	0.0384
L_BoardSize	0.0372	0.0386	0.0927	0.0833
L_IndependentDirectorRatio	0.0380	0.0446	0.0694	0.0700
L_Big4	0.1322	0.0602	0.2102	0.1092
L_LnAuditFee	0.1556	0.0572	0.2838	0.1072
L_AuditorTenure	0.0842	0.0461	0.1607	0.0610
L_IsChangeFirm	0.0805	0.0280	0.1226	0.0361

Notes: The table reports the mean and maximum absolute SMD across the five IMS quintile dose groups. Continuous controls are winsorized at the 1st and 99th percentiles in the complete baseline common sample before lag construction. "Before" denotes unweighted comparisons and "After" denotes comparisons using the original GPS weights. The 0.10 value is a conventional descriptive reference.

Table 7: Lagged Covariate and GPS–IPW Regressions

	(1) Lagged Covariates	(2) GPS–IPW	(3) GPS–IPW with Controls
IMS	-0.5162** (0.2515)	-0.8549*** (0.2737)	-0.2878** (0.1191)
Controls	Yes	No	Yes
Observations	10,488	10,488	10,488
Adjusted R^2	0.0244	0.0043	0.5654
Year fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes

Notes: The dependent variable is current Greenwashing. Column (1) reports the lagged-covariate specification with all baseline financial, governance, and audit controls measured in the preceding year. Columns (2) and (3) report the untrimmed GPS–IPW specifications; column (2) excludes outcome-equation controls and column (3) includes all baseline controls measured currently. All models include year and industry fixed effects and use firm-clustered standard errors. Coefficient estimates are reported first, followed by standard errors in parenthese.***, **, and * indicate two-tailed significance at 1, 5, and 10 percent.

Table 8: Auditor-Switching Selection Equation

	Logit model
Lagged Greenwashing coefficient	0.0852
	(0.0920)
p-value	0.3547
Controls	Yes
Observations	777
Year fixed effects	Yes
Industry fixed effects	Yes

Notes: The selection equation is estimated using Logit and includes the full set of lagged controls, together with lagged Greenwashing. Year and industry fixed effects are included, and standard errors are clustered by firm. Switching events require both $IsChangeFirm = 1$ and an observed change in InstitutionID in adjacent years. Coefficient estimates are reported first, followed by standard errors in parenthese.***, **, and * indicate two-tailed significance at 1, 5, and 10 percent

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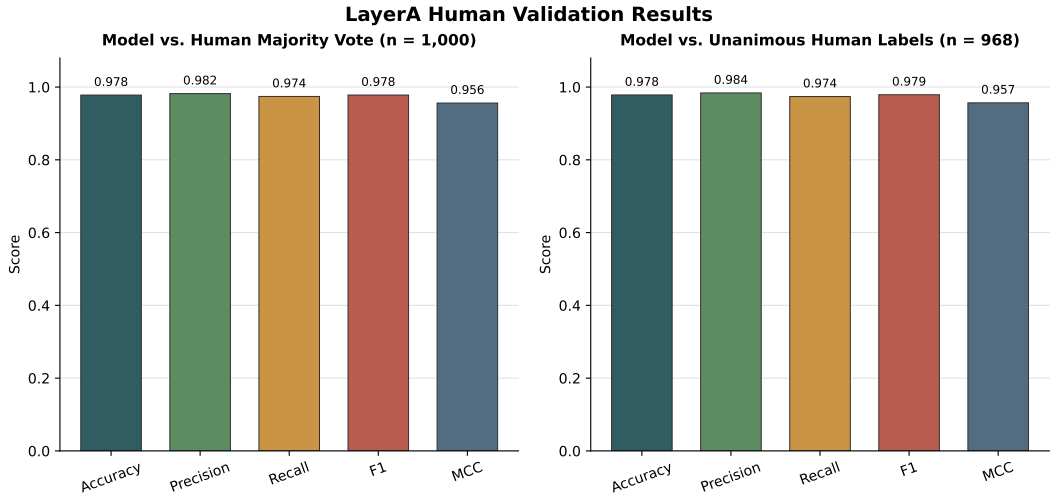
Zhou, Y., L. Chen, Y. Zhang, and W. Li. 2024. "Environmental disclosure greenwashing" and corporate value: The premium effect and premium devalue of environmental information. *Corporate Social Responsibility and Environmental Management* 31 (3): 2424–2438. https://doi.org/10.1002/csr.2698.

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A Semantic-Classification Validation

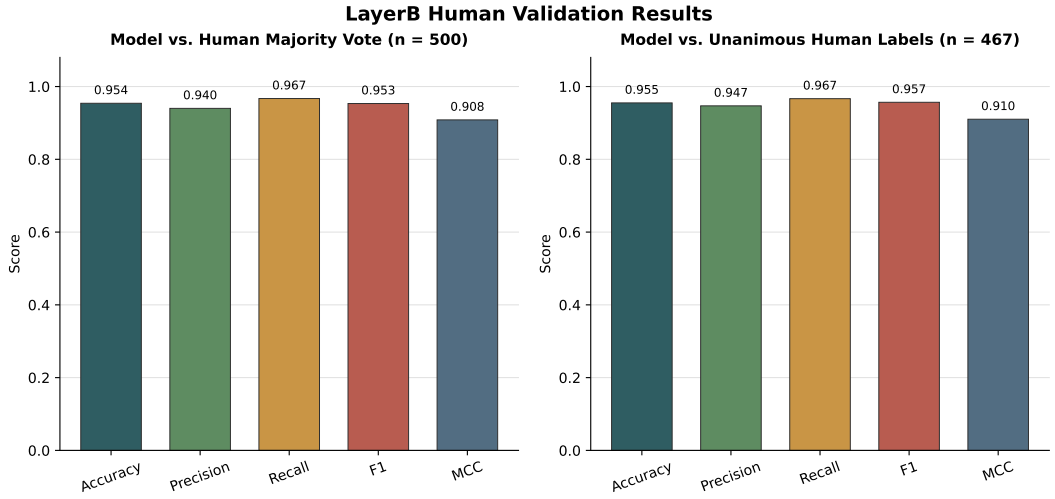
Adapting the human-validation approach in DeepGreen (Xu et al., 2026), we assess the reliability of the semantic classifications through independent human review. Three students with backgrounds in ESG and finance independently evaluated the same validation observations. The human reference label was determined by majority vote, with an observation assigned a value of one when at least two annotators assigned that label. The Layer A validation sample contains 1,000 observations, stratified into equal numbers of model-predicted positive and negative observations within each of ten review rounds. The Layer B validation sample contains 500 observations, balanced between model-predicted substantive and symbolic classifications.

Figures 1 and 2 report Accuracy, Precision, Recall, F1, and MCC relative to both the human majority label and the subset on which all three annotators agree, together with the annotators' pairwise Cohen's kappa statistics. Relative to the human majority label, Layer A achieves accuracy of 0.978 and an F1 score of 0.978, while Layer B achieves accuracy of 0.954 and an F1 score of 0.953. The annotators fully agree on 96.8 percent of Layer A observations and 93.4 percent of Layer B observations. These results indicate high agreement both between the model and human reviewers and among the reviewers themselves.



Human Annotator Pair	Cohen's kappa
Annotator 1 vs. Annotator 2	0.966
Annotator 1 vs. Annotator 3	0.970
Annotator 2 vs. Annotator 3	0.936

Figure 1: Layer A human-validation results. The bar charts compare model classifications with majority-vote and unanimous human labels; the table reports pairwise Cohen's kappa.



Human Annotator Pair	Cohen's kappa
Annotator 1 vs. Annotator 2	0.900
Annotator 1 vs. Annotator 3	0.968
Annotator 2 vs. Annotator 3	0.868

Figure 2: Layer B human-validation results. The bar charts compare model classifications with majority-vote and unanimous human labels; the table reports pairwise Cohen's kappa.

B Baseline Regression Robustness

The baseline regressions use firm-clustered standard errors. This appendix evaluates whether the inference is robust to alternative clustering choices. Industry–year two-way clustering allows dependence within industries and common shocks within years, while industry clustering provides a single-dimension benchmark that permits arbitrary within-industry dependence. The specifications retain the baseline sample, explanatory variables, controls, and year and industry fixed effects; only the standard-error estimator changes.

Table 9 reports the results. The coefficient on IMS is -0.4829 under all three clustering schemes. The standard errors and significance levels vary modestly, and IMS remains negative and statistically significant in each column. These results support the conclusion that the baseline association between audit-firm industry specialization and Greenwashing is not driven by the firm-clustered standard-error choice.

Table 9: Robustness of Baseline Estimates to Alternative Standard-Error Clustering

	(1) Firm-clustered SE	(2) Two-way clustered SE	(3) Industry-clustered SE
IMS	-0.4829**	-0.4829**	-0.4829*
	(0.2362)	(0.2348)	(0.2639)
Controls	Yes	Yes	Yes
Observations	15,381	15,381	15,381
Adjusted R^2	0.0221	0.0221	0.0221
Year fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes

Notes: The dependent variable is current Greenwashing. All columns estimate the baseline specification with the full set of financial, governance, and audit controls. Only the standard-error clustering changes across columns. Coefficient estimates are reported first, followed by standard errors in parentheses. ***, **, and * indicate two-tailed significance at 1, 5, and 10 percent.

C Additional GPS Specifications

C.1 GPS–IPW Construction

Because IMS lies between zero and one, we use its logit transformation,

$$Z_{i,t} = \text{logit}(IMS_{i,t}) = \ln\left(\frac{IMS_{i,t}}{1 - IMS_{i,t}}\right).$$

We then estimate the conditional treatment model

$$Z_{i,t} = \eta' X_{i,t-1} + \lambda_t + \delta_j + u_{i,t}, \qquad u_{i,t} \sim N(0, \sigma^2),$$

where $X_{i,t-1}$ contains the lagged outcome, financial, governance, and audit-selection variables, and λ_t and δ_j denote year and industry fixed effects. The estimated GPS is the conditional density

$$\widehat{r}_{i,t} = \widehat{f}_{Z|X}\left(Z_{i,t} \mid X_{i,t-1}, \lambda_t, \delta_j\right) = \phi\left(Z_{i,t}; \widehat{\mu}_{i,t}, \widehat{\sigma}\right),$$

where $\widehat{\mu}_{i,t}$ is the fitted value from the first-stage model and $\widehat{\sigma}$ is the residual standard deviation. The lagged panel therefore supplies information about observable determinants of current treatment assignment without conditioning the first stage on variables that may be current-treatment consequences.

The stabilized inverse probability weight is

$$w_{i,t} = \frac{\widehat{f}_Z(Z_{i,t})}{\widehat{f}_{Z|X}(Z_{i,t} \mid X_{i,t-1}, \lambda_t, \delta_j)},$$

where the denominator is the estimated GPS and $\widehat{f}_Z$ is the normal marginal density of Z, evaluated at the sample mean and sample standard deviation of the logit-transformed treatment (Robins et al., 2000; Naimi et al., 2014). Stabilization limits the influence of the marginal treatment distribution on the weighted sample, while the denominator adjusts for the observed assignment mechanism. The effective sample size is

$$ESS = \frac{\left(\sum_i w_{i,t}\right)^2}{\sum_i w_{i,t}^2}.$$

The main analysis uses the untrimmed weights.

To assess covariate balance, we divide IMS into quintile dose groups and compare each group with the remaining observations using standardized mean differences.[5] Table 6 reports, for every first-stage covariate, the mean and maximum absolute SMD across these five groups before and after weighting.

The weighted outcome model is

$$Greenwashing_{i,t} = \alpha + \beta IMS_{i,t} + \gamma' Controls_{i,t} + \lambda_t + \delta_j + \varepsilon_{i,t},$$

where $Controls_{i,t}$ contains all three baseline classes of current controls and standard errors are clustered by firm. Model (2) of Table 7 omits these controls, and Model (3) includes them.

C.2 Weight-Trimming Sensitivity

Table 10 reports weight-trimming sensitivity for the same specification. The only difference across rows is the treatment of the estimated GPS weights. All GPS first-stage covariates are constructed after applying the baseline 1st–99th percentile winsorization to continuous controls in the complete baseline common sample.

The untrimmed specification retains every estimated weight and therefore preserves the original GPS reweighted target population, but it can be more sensitive to observations with relatively low conditional treatment density and unusually large weights. P0.5–P99.5 and P1–P99 trimming limits the influence of these extreme weights and increases the effective sample size, but it also changes the weighting scheme and hence the exact target pseudo-population. The negative coefficient remains similar across the three weight treatments, providing supplementary evidence that the main association is not driven solely by the most influential weights.

[5] The five groups are the empirical quintiles of IMS: Q1, $(-0.0009871, 0.00997]$; Q2, $(0.00997, 0.0285]$; Q3, $(0.0285, 0.0608]$; Q4, $(0.0608, 0.114]$; and Q5, $(0.114, 0.731]$. The displayed lower bound of Q1 is a software-generated interval boundary; the observed IMS values are nonnegative. These quintile groups are used only to diagnose covariate balance and do not replace the continuous IMS treatment in the GPS estimation or outcome regression.

Table 10: GPS–IPW Weight-Trimming Sensitivity

Weight treatment	IMS	Standard error	p-value	ESS
Untrimmed	-0.2878	0.1191	0.0157	7,485
P0.5–P99.5 trimmed	-0.2351	0.1152	0.0413	8,529
P1–P99 trimmed	-0.2221	0.1147	0.0529	8,857

Notes: All models use 10,488 firm-years and 4,168 firms, include year and industry fixed effects, and use firm-clustered standard errors. ESS is the effective sample size. Weight trimming is applied to the estimated GPS weights.

C.3 Binary Complements

Because IMS is continuous, the GPS–IPW analysis retains variation in specialization intensity. As a complementary check on observable selection, we recast specialization as the binary indicator IMS_D and re-estimate the association using entropy balancing (Hainmueller, 2012) [6] and propensity-score matching (Minutti-Meza, 2013; Shipman et al., 2017)[7]. Table 11 reports the results. Column (1) uses entropy balancing and column (2) uses propensity-score matching. In both columns the coefficient remains negative and statistically significant, consistent with the GPS–IPW estimates.

Table 11: Binary Complements to GPS–IPW

	(1) Entropy Balancing	(2) Propensity-Score Matching
IMS	-0.3406** (0.1542)	—
IMS_D	—	-0.1120** (0.0534)
Controls	Yes	Yes
Observations	10,488	1,218
Adjusted R^2	0.6079	0.5906
Year fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes

Notes: The dependent variable is current Greenwashing. Column (1) reports the entropy-balancing specification and column (2) reports the propensity-score matching specification. Coefficient estimates are reported first, followed by standard errors in parentheses. All models include year and industry fixed effects and use firm-clustered standard errors. Column (1) uses 10,488 firm-years from 4,168 firms; the effective sample size is 2,245. Column (2) uses 1,218 matched firm-years from 792 firms (609 pairs). ***, **, and * indicate two-tailed significance at 1, 5, and 10 percent.

[6] We implement Hainmueller's (2012) ATT entropy balancing by reweighting untreated observations so that the first moments of the lagged financial, governance, and audit-selection covariates, and the second moments of the continuous lagged covariates, match the treated group.

[7] Following Minutti-Meza (2013) and Shipman et al. (2017), we estimate a logit propensity-score model with the full set of lagged covariates, impose common support, and match treated and control observations 1:1 without replacement using a caliper of 0.03.